A. Hyperparameters

This section details the key hyperparameters used in THE AI SCIENTIST-V2. Model configurations for language and vision-language models are listed in Table 2. The hyperparameters governing the agentic tree search (§3.2.2) and experiment stage (§3.2.1) progression, including node execution limits, are shown in Table 3.

Table 2 | LLM and VLM Hyperparameters.

Component/Task	Model Used	Max Tokens	Temperature
Code Generation (§3.2)	Claude 3.5 Sonnet (v2)	8,192	0.5
LLM/VLM Feedback Agents (§3.4)	GPT-4o	8,192	0.5
Summary Report Agent (§3)	GPT-4o	8,192	1.0

Table 3 | Agentic Tree Search & Execution Hyperparameters (§3.2.2, §3.2.1).

Hyperparameter	Value
Debug Probability	1.0
Maximum Debug Depth	3
Maximum Experiment Runtime per Node	1 hour
Node Allocation per Stage:	
Stage 1: Preliminary Investigation	21 nodes
Stage 2: Hyperparameter Tuning	12 nodes
Stage 3: Research Agenda Execution	12 nodes
Stage 4: Ablation Studies	12 nodes

The total time required for THE AI SCIENTIST-V2 to generate a single paper depends on the complexity of the problems. Based on our experience, this process usually takes anywhere from several hours to a maximum of 15 hours, which is the runtime limit we have set.

B. Prompts

In this section, we include the prompts used in all phases of THE AI SCIENTIST-V2.

Idea Generation Prompt

```
# System prompt
You are an experienced AI researcher who aims to propose high-impact
research ideas resembling exciting grant proposals. Feel free to propose
any novel ideas or experiments; make sure they are novel. Be very creative
and think out of the box. Each proposal should stem from a simple and
elegant question, observation, or hypothesis about the topic. For example,
they could involve very interesting and simple interventions or
investigations that explore new possibilities or challenge existing
assumptions. Clearly clarify how the proposal distinguishes from
the existing literature.

Ensure that the proposal can be done starting from the provided
codebase, and does not require resources beyond what an academic
lab could afford. These proposals should lead to papers that are
publishable at top ML conferences.
```